PCA & t-SNE: Ultra-Deep Visuals

20 Advanced Charts for the Curious Data Scientist

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Methods and Algorithms — MSc Data Science

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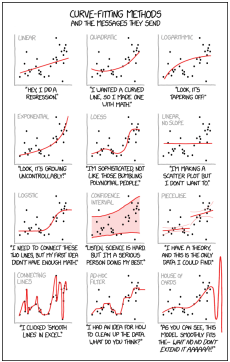
The Art of Fitting



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Learning Objectives

After this lecture, you will be able to:

- **Diagnose** PCA anomalies using Hotelling T^2 and SPE statistics
- **Evaluate** t-SNE embedding quality via trustworthiness and KL divergence diagnostics
- **Compare** dimensionality reduction methods for suitability on different data structures
- **Apply** PCA/t-SNE to realistic finance scenarios (yield curves, portfolios, credit risk)

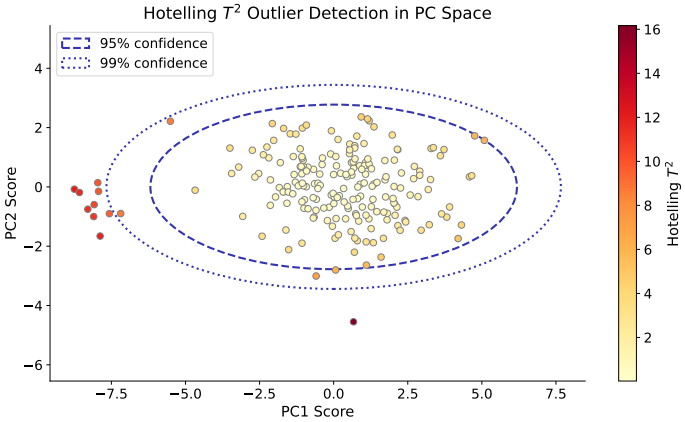
We cover 20 advanced visualizations spanning PCA diagnostics, t-SNE mechanics, practical pipelines, and finance applications

Outline

1. PCA Deep Mechanics
2. t-SNE Deep Mechanics
3. Practical Diagnostics
4. Finance Applications
5. Summary

Four parts: PCA Deep Mechanics, t-SNE Deep Mechanics, Practical Diagnostics, Finance Applications

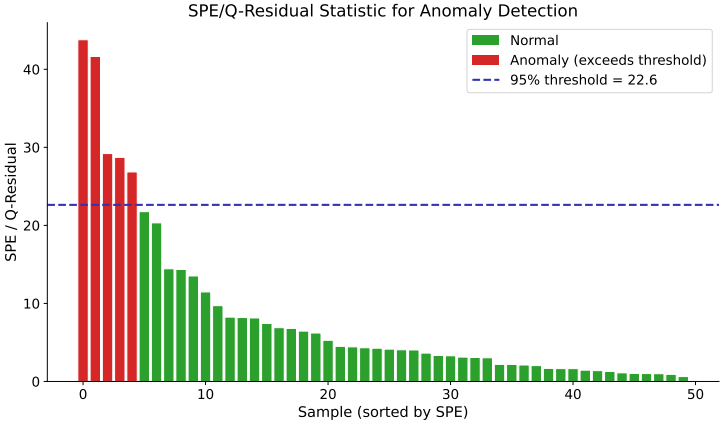
Can PCA Detect Outliers? The Hotelling T^2 Statistic



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_21_hotelling_t2_outlier

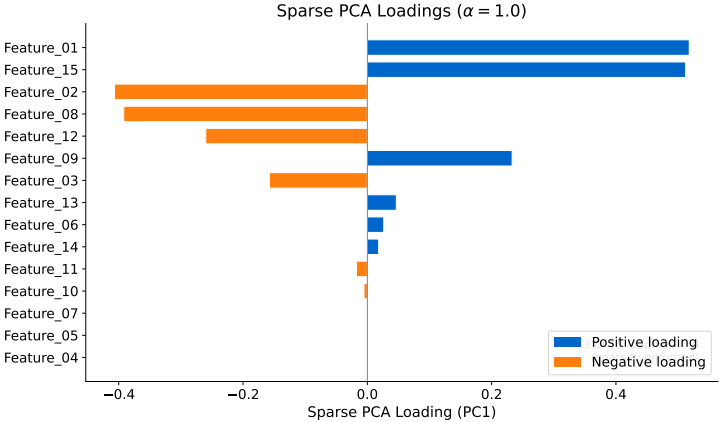
Hotelling T^2 measures Mahalanobis distance in PC space—points beyond the 99% ellipse are statistically anomalous

What PCA Misses: The Q-Residual (SPE) Statistic



SPE detects anomalies invisible in PC space—deviations in directions PCA discards as noise

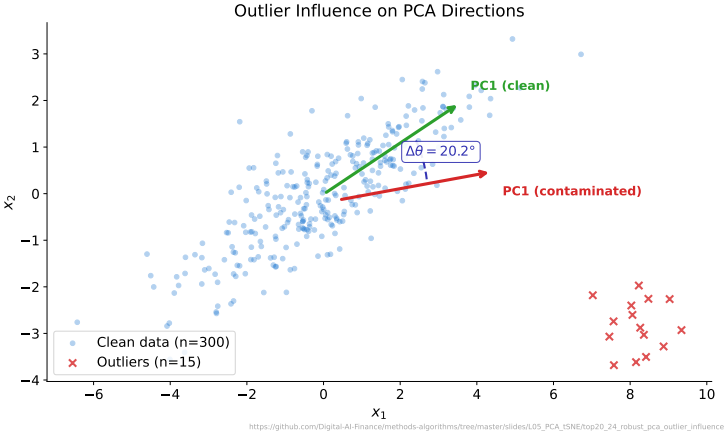
Sparse PCA: Can We Get Interpretable Components?



https://github.com/Digital-AI-finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_23_sparse_pca_loadings

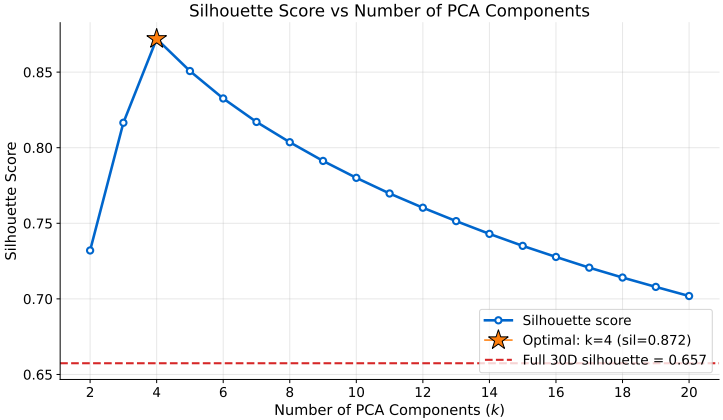
Sparse PCA trades variance explained for interpretability—each component loads on only a handful of features

How Sensitive Is PCA to Outliers?



Just 5% contamination can rotate PC1 by 20+ degrees—always inspect for outliers before running PCA

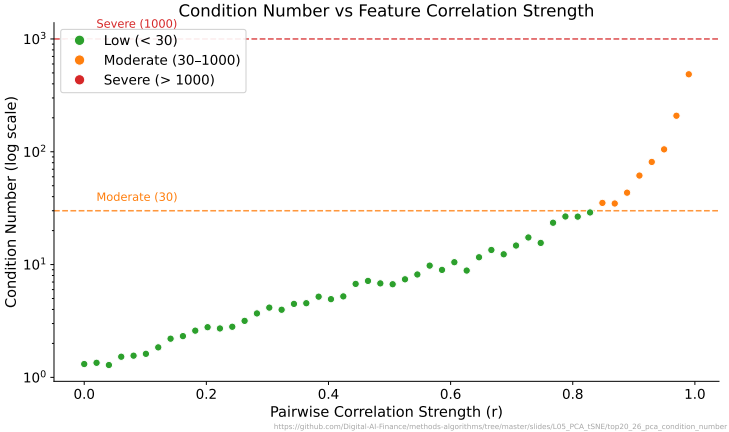
How Does the Number of Components Affect Cluster Separability?



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_25_pca_variance_vs_components

Removing noise dimensions with PCA can improve clustering quality—reduced data sometimes has higher silhouette than the original

Why Does Multicollinearity Break PCA?



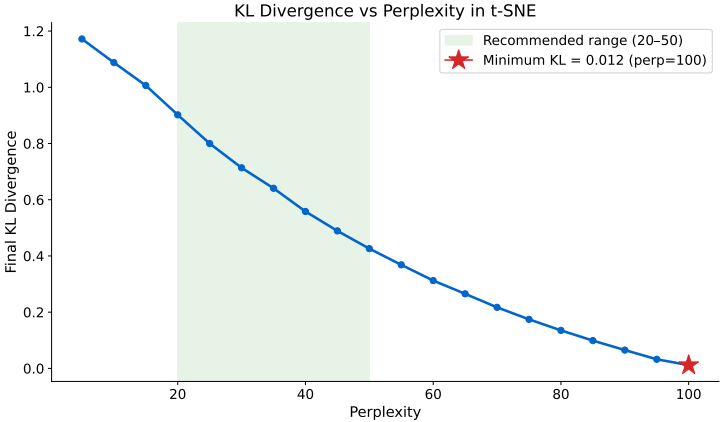
High correlation produces near-singular covariance matrices—PCA becomes numerically unstable with condition numbers above 1000

From PCA Mechanics to t-SNE

- PCA captures **global linear structure** but fails on curved manifolds
- t-SNE preserves **local neighbourhood structure** in low dimensions
- The next 6 charts explore t-SNE's inner mechanics

Understanding t-SNE's hyperparameters is essential for producing reliable embeddings

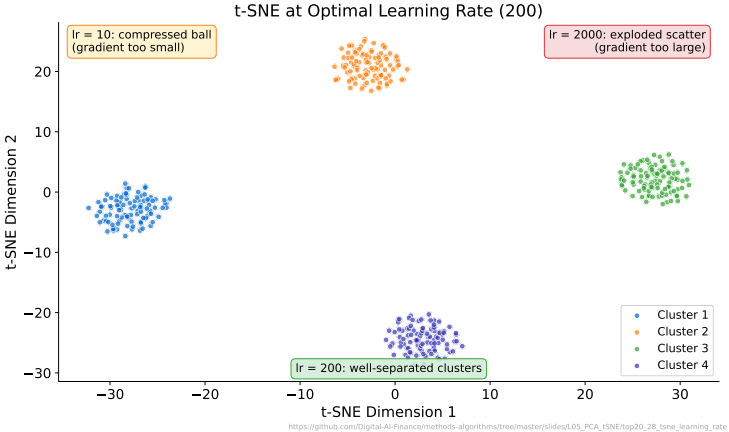
Which Perplexity Gives the Best Embedding?



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_27_tsne_perplexity_vs_kl

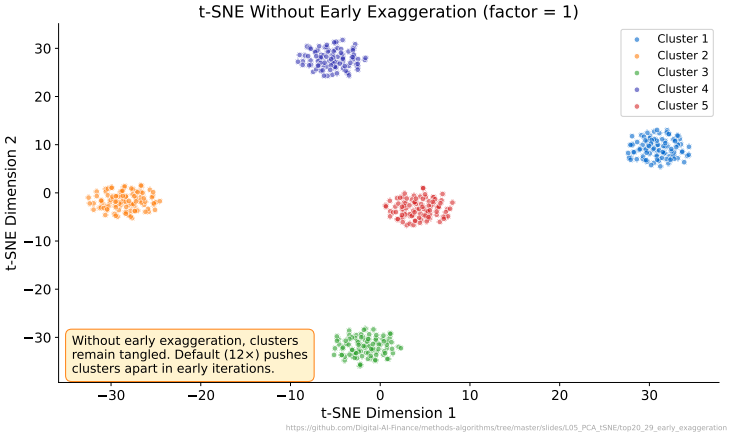
KL divergence quantifies embedding quality—sweep perplexity values and pick the minimum for your dataset

What Happens When You Change the t-SNE Learning Rate?



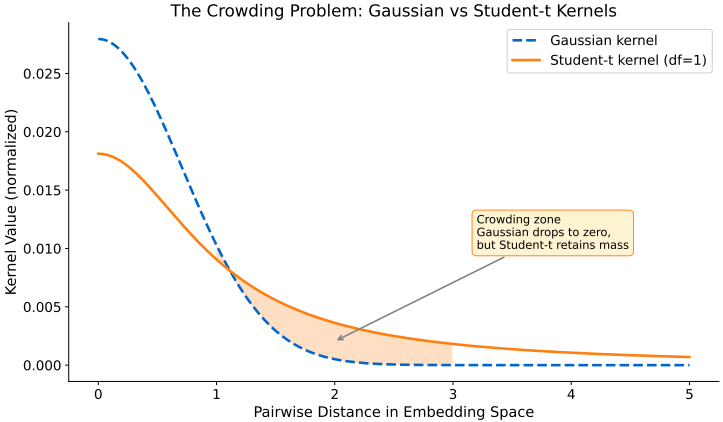
Learning rate too low: compressed blob. Too high: uniform scatter. Default (N/early_exaggeration or 200) is usually right

Why Does t-SNE Use Early Exaggeration?



Early exaggeration (default 12x) forces clusters apart in the first 250 iterations—without it, clusters remain entangled

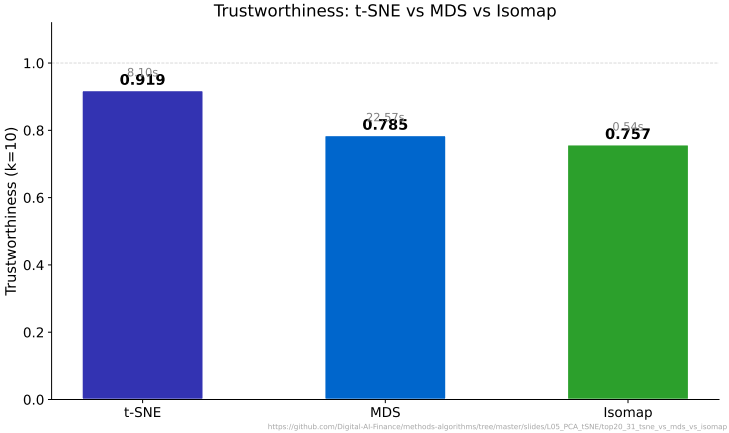
The Crowding Problem: Why t-SNE Uses a t-Distribution



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_30_crowding_problem

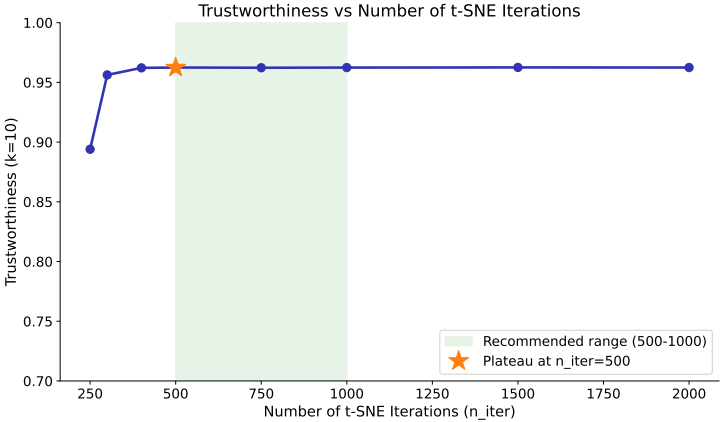
The t-distribution's heavy tails give moderately distant points more room—this is the key insight that makes t-SNE superior to SNE

t-SNE vs MDS vs Isomap: Which Preserves What?



t-SNE preserves local structure best; MDS preserves global distances; Isomap follows geodesic paths along the manifold

How Many Iterations Does t-SNE Actually Need?



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_32_tsne_n_iter_effect

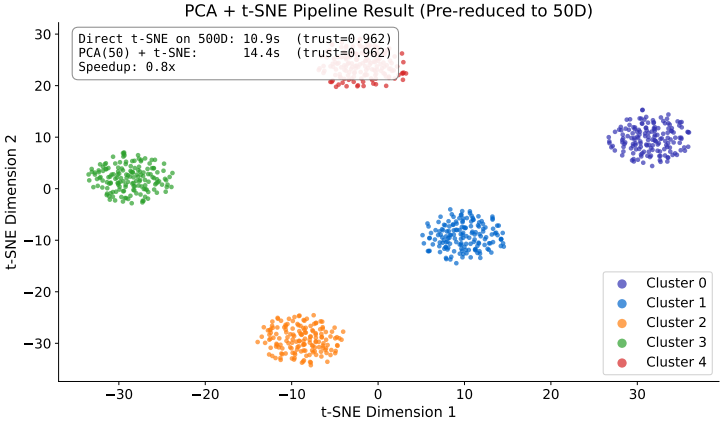
t-SNE quality plateaus around 500–1000 iterations—the default 1000 is usually sufficient, but always verify

From Theory to Practice

- Combine PCA and t-SNE in **pipelines** for scalable embeddings
- Measure what "good" means with **formal metrics**
- Bridge the gap between variance explained and task performance

Practical diagnostics turn dimensionality reduction from art into engineering

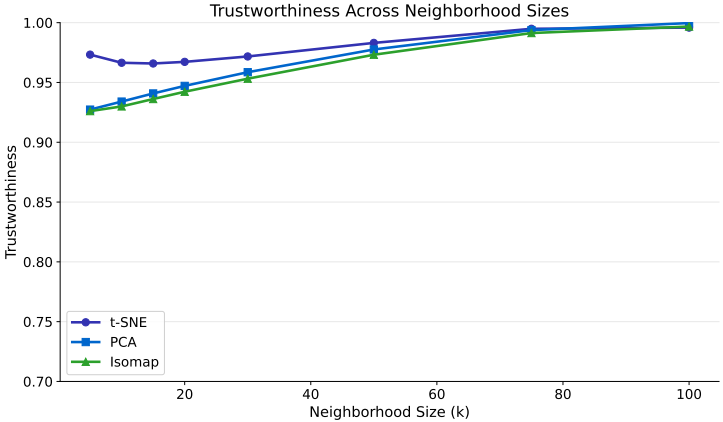
The PCA-then-t-SNE Pipeline: Why Pre-Reduce to 50D?



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_33_pca_then_tsne_pipeline

Pre-reducing to 50D with PCA removes noise dimensions and can speed up t-SNE by 10x or more

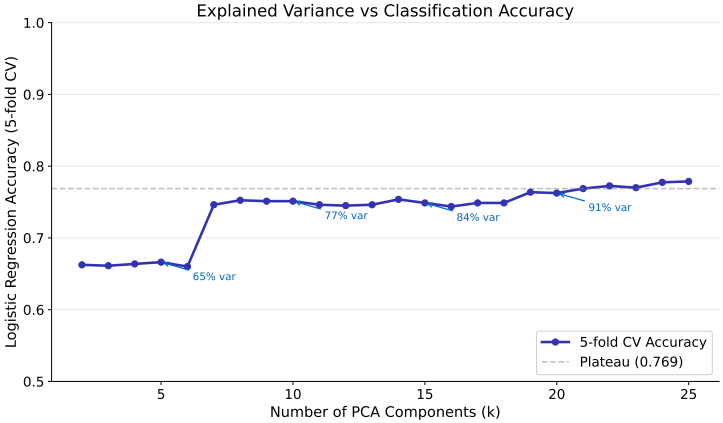
How Does Neighbourhood Size Affect Embedding Quality?



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_34_trustworthiness_vs_k

Trustworthiness at small k measures local fidelity; at large k it measures global fidelity—t-SNE excels locally, PCA globally

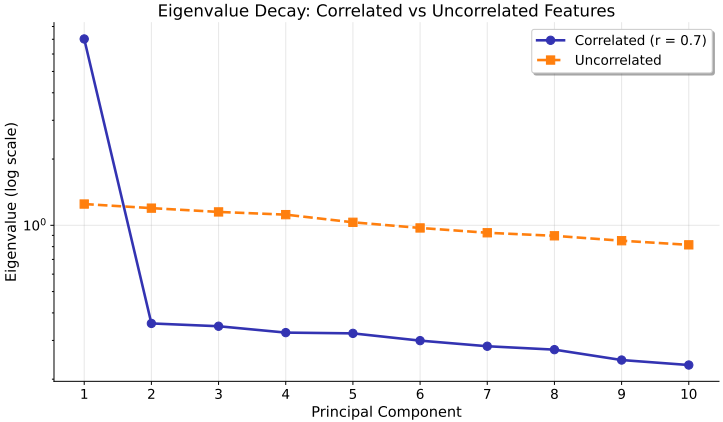
More Variance Explained = Better Predictions?



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_35_explained_var_vs_accuracy

The 95% variance rule is a useful heuristic but not gospel—downstream accuracy often saturates much earlier

Why Does Correlation Determine PCA's Usefulness?



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_36_pca_correlation_effect

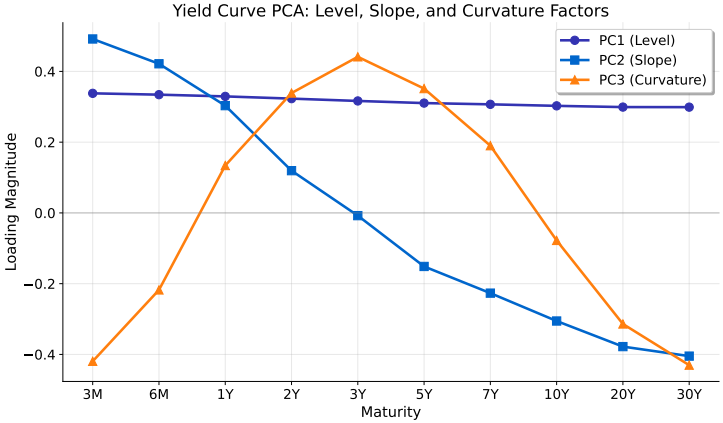
PCA compresses correlated data efficiently—uncorrelated features yield flat eigenvalues where no component can be dropped

From Theory to Finance

- PCA reveals **hidden risk factors** in bond and equity markets
- t-SNE uncovers **credit risk structure** invisible to linear methods
- Rolling PCA monitors **portfolio diversification** in real time

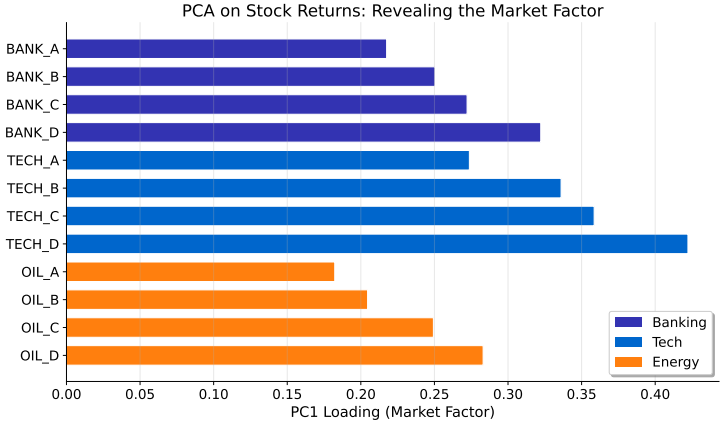
Dimensionality reduction is a workhorse tool in quantitative finance

What Do Yield Curve PCA Factors Actually Look Like?



PCA on yield curves reveals Level, Slope, and Curvature—three factors that explain 98% of interest rate movements

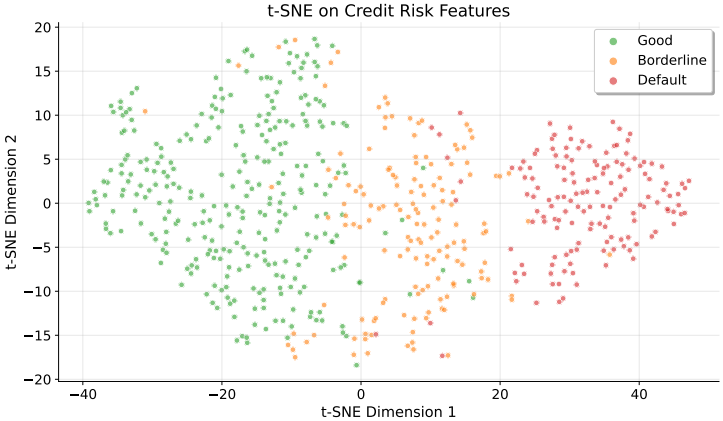
Can PCA Reveal Hidden Factors in a Stock Portfolio?



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_38_portfolio_factor_decomposition

PCA on returns reveals the market factor (PC1) and sector rotations—the empirical foundation of factor investing

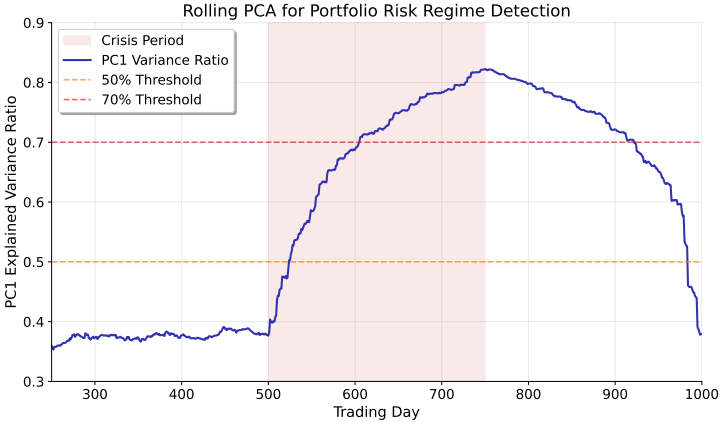
Can t-SNE Reveal Structure in Credit Risk Features?



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_39_credit_risk_tsne

t-SNE on credit features reveals natural risk clusters—the borderline region shows where classification is genuinely difficult

PCA for Portfolio Risk Monitoring: Detecting Regime Changes



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L05_PCA_tSNE/top20_40_portfolio_risk_pca_monitoring

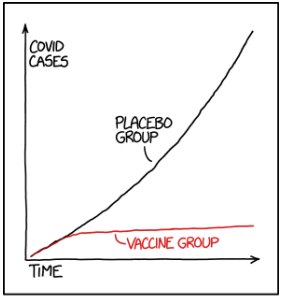
When PC1's share spikes above 70%, correlations are rising and diversification is breaking down—a crisis early-warning signal

What We Covered: 20 Visuals Checklist

- ✓ Hotelling T^2 Outlier Detection
- ✓ Q-Residual (SPE) Statistic
- ✓ Sparse PCA Loadings
- ✓ Robust PCA vs Outliers
- ✓ Variance vs Components
- ✓ PCA Condition Number
- ✓ Perplexity vs KL Divergence
- ✓ t-SNE Learning Rate
- ✓ Early Exaggeration
- ✓ Crowding Problem
- ✓ t-SNE vs MDS vs Isomap
- ✓ t-SNE Iteration Effect
- ✓ PCA-then-t-SNE Pipeline
- ✓ Trustworthiness vs k
- ✓ Variance vs Accuracy
- ✓ Correlation Effect on PCA
- ✓ Yield Curve Factors
- ✓ Portfolio Factor Decomposition
- ✓ Credit Risk t-SNE
- ✓ Portfolio Risk Monitoring

These 20 advanced visualizations cover PCA diagnostics, t-SNE mechanics, practical pipelines, and finance applications

Never Stop Exploring Your Data



STATISTICS TIP: ALWAYS TRY TO GET DATA THAT'S GOOD ENOUGH THAT YOU DON'T NEED TO DO STATISTICS ON IT

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